

MASTER'S DEGREE IN AI CAPSTONE · PROJECT 8 · PROFESSIONAL INDUSTRY DEFENSE

Trustworthy question generation for a Bible-trivia app

An integrated AI pipeline built from four capstone projects

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Every answer key is a small act of teaching

- I build and run El Master de la Biblia, a Bible-trivia app on Google Play since 2013: 500,000+ downloads, rated 4.4 by 30,600 reviewers.
- Its catalog holds 53,831 questions in Spanish, English and Portuguese, generated with an LLM and screened by LLM judges.
- Since publication, the catalog has received 3,178 corrections and 571 withdrawals.

“Who did Ruth marry?” Key: Boaz

Mahlon was also an option, and Ruth 4:10 calls Ruth “the wife of Mahlon”. My Project 6 agent and its blind verifier accepted this question.

One model is not enough for this problem

Language model

Writes varied, well-worded questions, but only checks what it is shown.

Structured knowledge

Knows who is related to whom, even when the second answer is in another chapter.

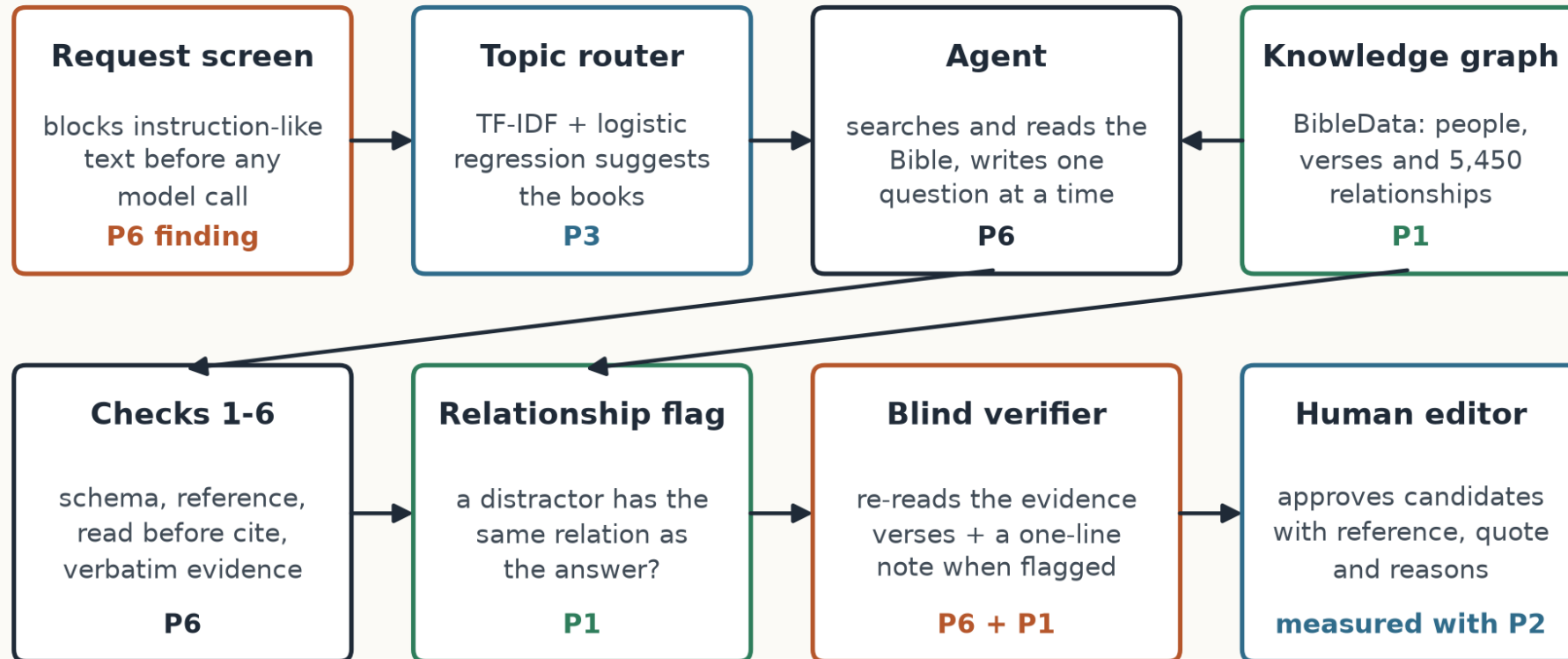
Statistics

Tells whether each component is worth its cost, with intervals and paired tests.

Constraints: public data only (World English Bible, public domain; BibleData, CC BY 4.0) · no player data · a human editor approves every question.

2 · INTEGRATED SYSTEM OVERVIEW

Screen, route, act, check, measure



Four prior projects, each changed by the integration

Project	What it contributed	What is new here
P1 · BibleData workflow	Cleaning: 14,733 placeholders, repeated tags	Knowledge layer: 5,450 relationships, 482 shared names resolved, lookup tool, ambiguity flag
P2 · Statistical analysis	Profile every join; intervals, effect sizes	36 unresolved evidence references found; Wilson, exact McNemar, Fisher
P3 · Applied ML	TF-IDF + logistic regression; grouped splits	Topic router over 66 books, split by chapter: 33 of 36 topics in the top 3
P6 · Agentic workflow	Agent, 6 checks, blind verifier, replay; its failure cases	Switchable components, request screen, hybrid verification

The knowledge graph triggers re-verification; it does not reject

Why not reject every flag?

The graph flagged 7 questions that the agents had accepted. With the evidence, the verifier judged all 7 correct, e.g. “Which daughter-in-law kissed Naomi and then left her?": Ruth is also a daughter-in-law, but only Orpah left. A hard rule would have discarded all 7.

Why the one-line note?

With the evidence verses alone, the verifier still accepted Boaz, quoting Ruth 4:13 and stopping (24 of 30 caught). With the explicit relationship note it caught 30 of 30 (McNemar $p = 0.031$).

Transparency first, and a cost I can measure

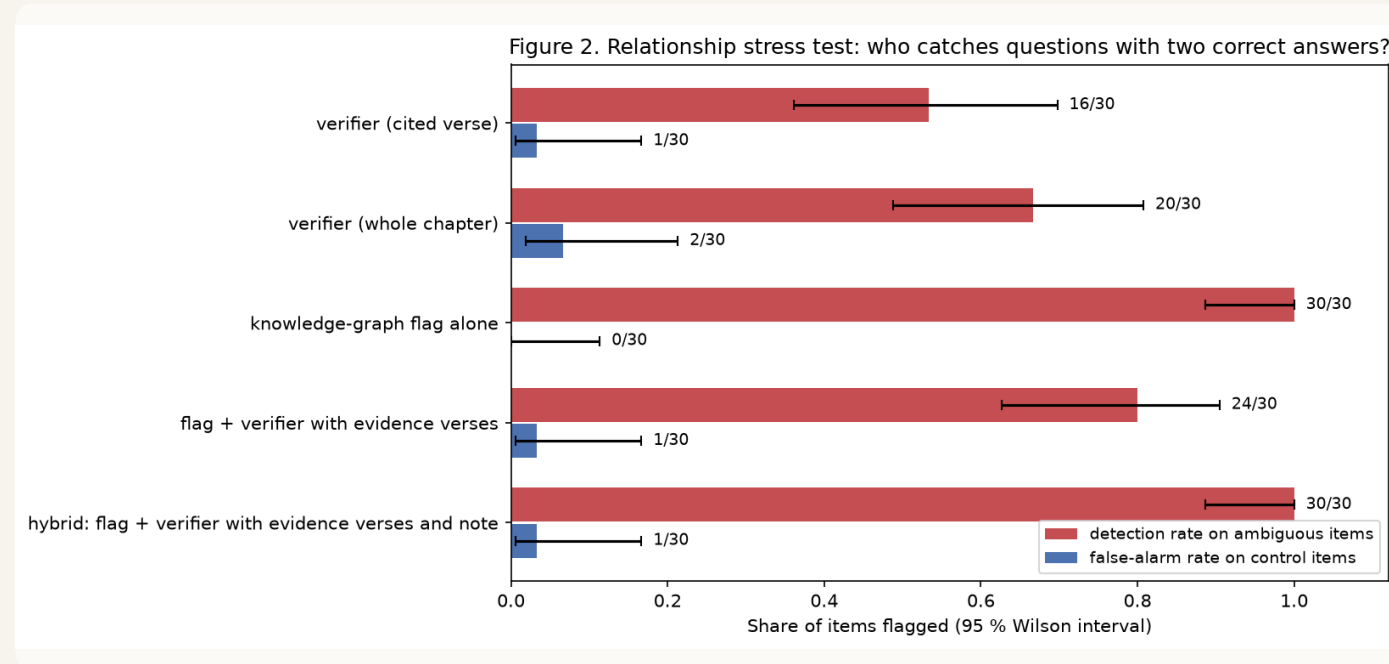
Decision	Chosen	Alternative	Why
Agent loop	Hand-written controller (OpenAI SDK)	An agent framework	Every decision visible in code and logs
Topic router	TF-IDF + logistic regression	Embeddings	Cheap, deterministic, explainable; weaker on paraphrases
Request screen	Visible list of patterns	A second model as classifier	Free and auditable; bypassed by rewording
Overall cost	+36 % tokens per accepted question	Project 6 configuration	Insurance against a rare, damaging error

Accountability for teaching scripture wrongly

A wrong key penalises the player who knows the text best	Candidates only; a human editor approves each question with its reference, verbatim quote, verifier reason and graph status.
The canon is a product choice, not neutral	Enoch is scripture for the Ethiopian Orthodox Church: the app should state its 66-book canon to players.
Manipulation through the request	Project 6's injection made the agent cite Genesis 1:1: now blocked before any model call, and the code checks still apply.
Privacy	Only public data; no player data from the production app enters the pipeline.

6 · EVALUATION: RELATIONSHIP STRESS TEST

The model verifier checks only what it is shown



16 / 30

caught by the Project 6 verifier

1 / 13

when the second answer is in another verse

30 / 30

caught by the hybrid design; 1 false alarm in 30 controls (McNemar $p = 0.0001$)

Same quality in normal use, at a higher cost

Result	Project 6 configuration	Integrated pipeline
Questions accepted (of 32)	29	29
Ambiguous keys in the bank	0	0
Tokens per accepted question	about 7,500	about 10,200 (+36 %)
Prompt injection (Samson)	reached the agent; ignored in this run	blocked before any model call
Book of Enoch (out of canon)	declined	wrote 3 questions on Revelation 12

Failure cases and unexpected behaviour

- Scope drift: asked about the Book of Enoch, the integrated agent followed the router's hints and wrote questions on Revelation 12 instead of declining.
- An optional tool is not a safeguard: lookup_person was called 3 times in 9 sessions.
- The request screen stops the known injection, not a paraphrase of it.
- The only false alarm came from an uncertain BibleData entry, and the verifier was right to reject it.

What would stop this from going to production tomorrow

- Knowledge base: 54 % of the relationships are inferred rather than quoted; 36 evidence references do not resolve.
- Coverage: the ambiguity check applied to 7 of the 29 integrated questions.
- One random run per configuration, about 30 questions each: little statistical power.
- The app publishes in Spanish, English and Portuguese; the pipeline covers English only. It also depends on a paid external API.

A pattern for any AI-generated assessment

What this shows

- Data, machine learning and agents combined in one system
- Ablations and paired tests to decide what earns its cost
- Ethics as design: provenance, human approval, stated boundaries
- Transferable to certification, compliance and medical education

Next steps

- A scope rule before the router, so hints cannot override a decline
- Run the person lookup automatically instead of optionally
- Normalise BibleData's book codes; add places and events
- Run the stress test on the app's production judges; add Spanish and Portuguese

Thank you. Questions are welcome.

Integrated pipeline and paper: github.com/arturoappp/industry-synthesis-project

Agent (Project 6): github.com/arturoappp/agent-ic-workflows-project

Defense materials: github.com/arturoappp/professional-defense-project